

THC block-encoding Toffoli comparison: three constructions

Parameters: $N_{\text{up}} = 4$, $N_{\text{down}} = 22$, N_{IP} (central bond) = 256
 $N_k = k^3$ for k in $[1, 2, 3, 4, 5, 6]$ (so N_k in $[1, 8, 27, 64, 125, 216]$)
phase_bitsize $b = 32$

All three encode the periodic THC tensor (arXiv:2601.16379 Eq. 9):

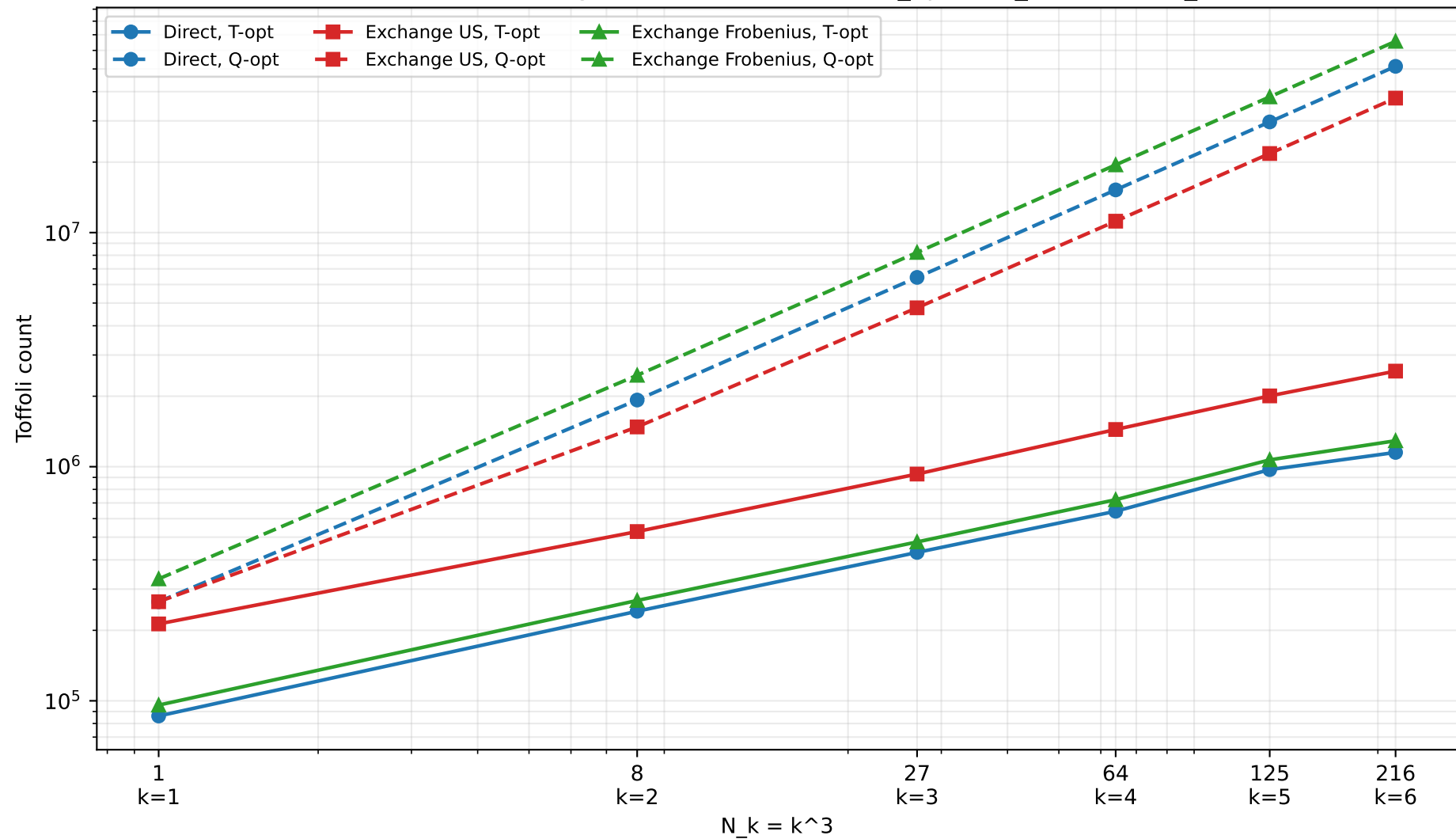
1. Direct: orientation flipped -- inputs $(\mu, k_\mu), (\lambda, k_\lambda)$, outputs the other two. $|Q\rangle$ uniform LCU, two X reflections, two momentum modular adds $k \pm Q$, central DIAGONAL kernel $\sum W^Q_{\{IJ\}} |QIJ\rangle\langle QIJ|$ (QROAM $\rightarrow$ Ry $\rightarrow$ QROAM †), then the two X † .
2. Exchange US (Unitary Synthesis): bra-pair X reflections + central W^q via the SVD block-unitary interferometer (default central encoder).
3. Exchange Frobenius (use_fro_BE=True): same outer X reflections, but the central W^q tensor (just W , not X) is encoded by the Clader-Frobenius block-matrix scheme (BlockDiagonalClassicalMatrixBlockEncoding, $U_A = U_L^\dagger U_R$).

Tuning policies (per curve):

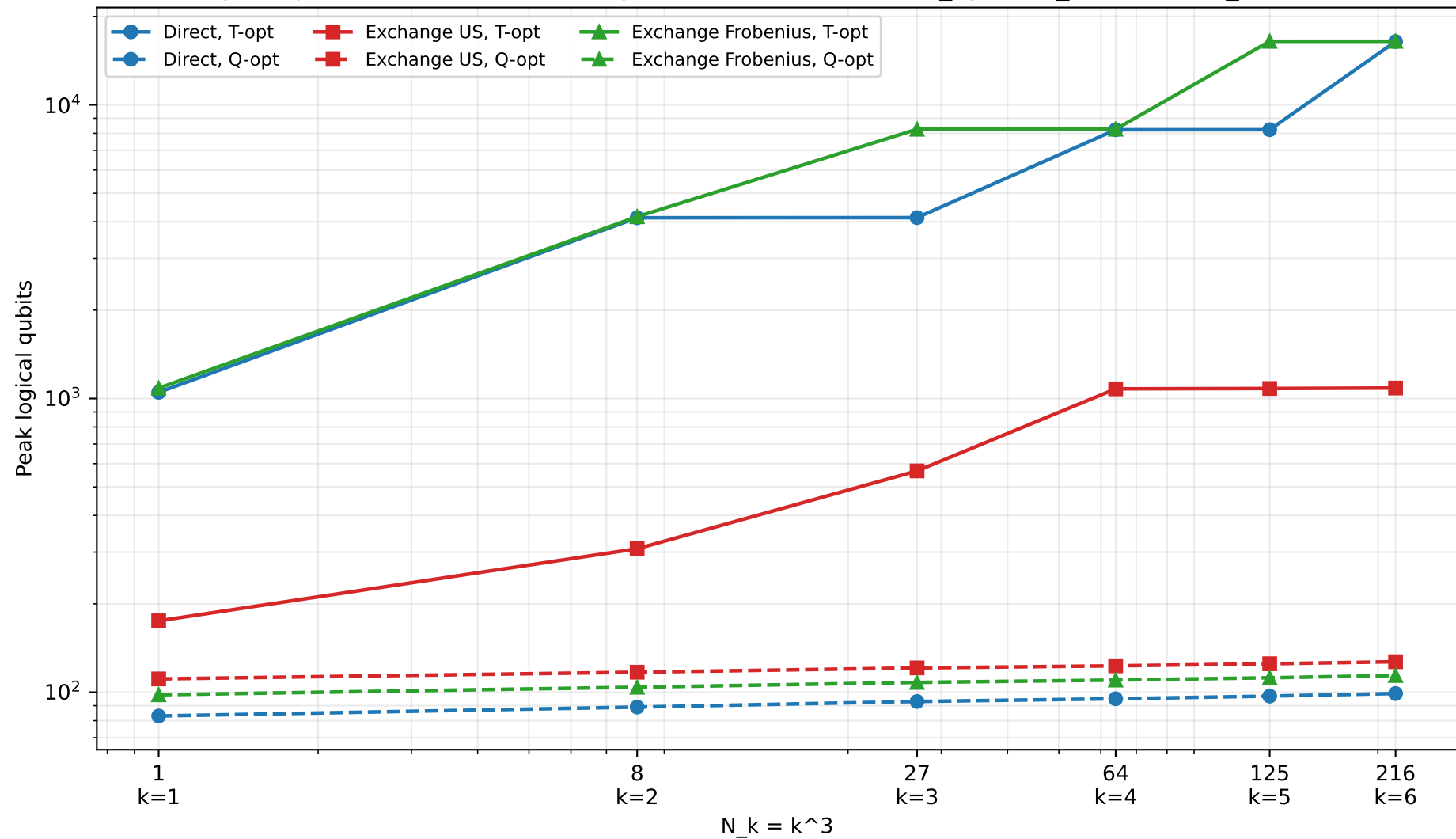
- T-opt: every QROAM log_block_sizes at the closed-form Toffoli optimum.
- Q-opt: no QROAM blocking (log_block_sizes = 0) $\rightarrow$ minimal qubits.

All Toffoli / qubit counts come from Qualtran's QECGatesCost / QubitCount walking each bloq's build_call_graph (no analytic formulas).

Toffoli count: THC block encoding, three constructions ($N_{\text{up}}=4$, $N_{\text{down}}=22$, $N_{\text{IP}}=256$, $b=32$)



Peak logical qubits: THC block encoding, three constructions ($N_{\text{up}}=4$, $N_{\text{down}}=22$, $N_{\text{IP}}=256$, $b=32$)



Resource counts: THC block encoding, three constructions

k	N_k	Direct T-opt T	Direct Q-opt T	Exch-US T-opt T	Exch-US Q-opt T	Exch-Fro T-opt T	Exch-Fro Q-opt T	Direct T-opt Q	Exch-US T-opt Q	Exch-Fro T-opt Q
1	1	86,130	264,923	213,032	264,973	95,863	331,734	1050	175	1085
2	8	241,310	1,926,447	528,414	1,478,501	268,303	2,457,410	4126	308	4161
3	27	430,334	6,436,319	930,190	4,772,377	477,031	8,227,119	4130	567	8260
4	64	645,164	15,218,613	1,441,878	11,186,673	722,596	19,462,774	8227	1079	8262
5	125	971,550	29,697,615	2,006,410	21,761,677	1,069,631	37,986,505	8229	1082	16455
6	216	1,150,724	51,297,365	2,560,306	37,537,429	1,290,669	65,620,194	16422	1086	16457